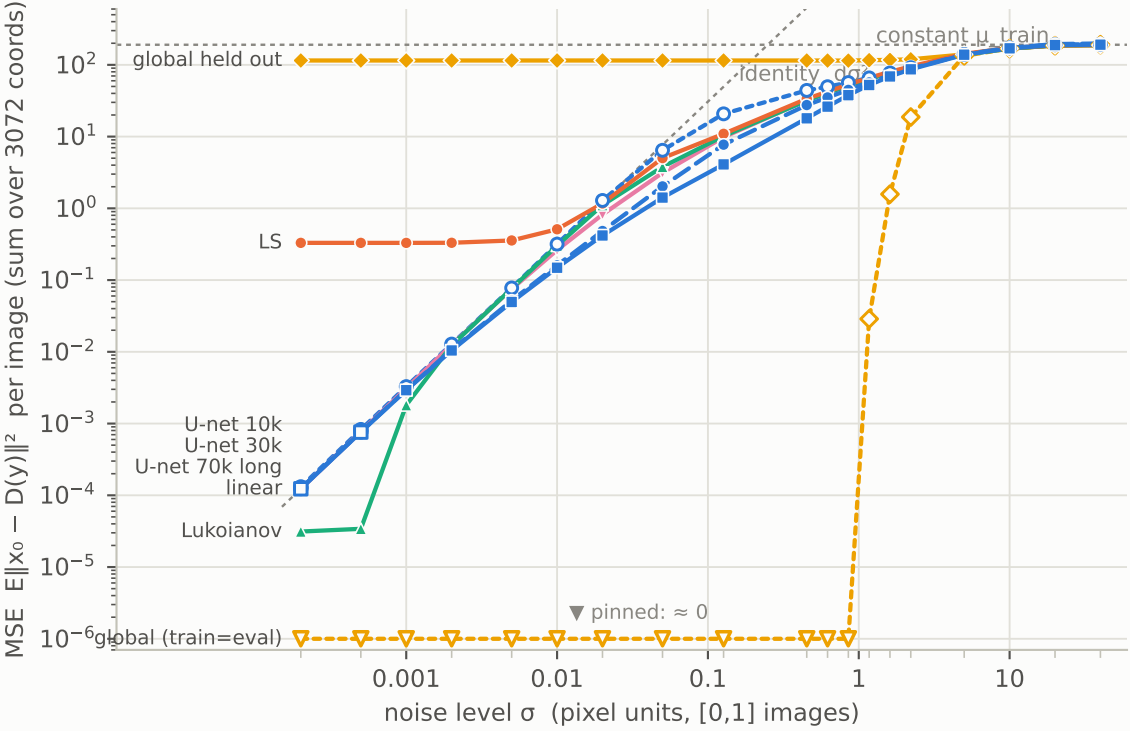
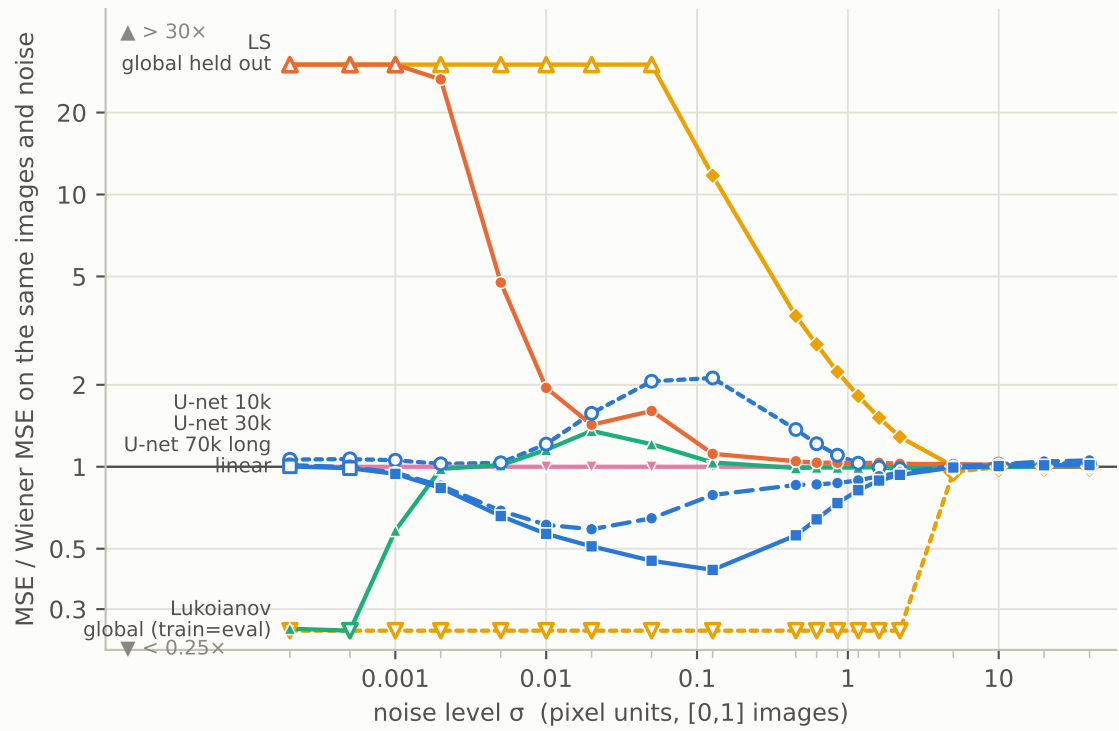


Held out: pool/fit = images [0:10000], test = [60000:61000], k and τ picked on the last 200 images. U-nets: same SongUNet; hollow = $\sigma < 0.001$ (below training range). train = eval: pool images [0:1000], pool contains each.

MSE vs noise, FFHQ 32x32



Relative to linear (< 1 beats Wiener)



Linear (Wiener) Softmax, global — held out Softmax, local — Kamb & Ganguli LS U-net, 30k [0:30k], held out
Softmax, global — train = eval Softmax, local — Lukoianov (Wiener mask) U-net, 10k = the pool, held out U-net, all 70k, 250k steps (test seen)

Lukoianov at $\sigma \leq 0.001 \rightarrow 0$ is an 8-bit artefact: as $\sigma \rightarrow 0$ its Wiener mask shrinks to the diagonal, a per-coordinate nearest value among 10k training pixels, which snaps to the exact 1/255 level once $\sigma \ll 1/255$. val-selected, by σ ascending: LS k = 1, 1, 1, 1, 1, 1, 1, 1, 3, 5, 7, 9, 11, 15, 19, 31, 31, 31, 31; Lukoianov τ = 0.005, 0.01, 0.05, 0.2, 0.5, 1, 0.3, 0.2, 0.1, 0.1, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.01, 0.01, 0.02